



GreenTravel Intelligence Challenge

A Hybrid EDA + Predictive ML Sustainability Hackathon

Official Participant Handbook

Welcome

"The greenest trip is the one you optimise before it happens."

Welcome to the GreenTravel Intelligence Challenge - a hackathon where data science meets corporate sustainability. Your mission: uncover hidden patterns in business travel data and build machine-learning models that can **predict high-carbon trips before they're booked.**

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1 The Sustainability Imperative

Business travel accounts for a staggering share of corporate carbon footprints. According to the Global Business Travel Association, air travel alone generates roughly 2–3% of global CO₂ emissions, and a single transatlantic business-class flight can emit over 2,000 kg of CO₂e per passenger - the equivalent of driving a car for six months.

As organisations race toward net-zero commitments, travel is one of the last remaining "hard to decarbonise" operational categories. Unlike electricity or fleet vehicles, travel decisions are deeply human - shaped by urgency, convenience, policy, and culture.

The opportunity: If we can predict which planned trips will result in disproportionately high emissions, we can intervene before booking - suggesting alternative transport modes, routing, or virtual meeting options. s

2 Challenge Overview

The GreenTravel Intelligence Challenge consists of two parts. **Participants must complete both parts.**

Component	Description
Focus	Binary Classification
Objective	Predict whether a business trip will be classified as a high-carbon trip before it is booked.
Tools	Python and standard machine learning libraries such as scikit-learn, XGBoost, LightGBM, CatBoost, or similar frameworks.
Input Data	Public trip dataset together with the provided event logs and attribute tables for feature engineering.
Output	A prediction CSV containing probability scores for the private test set, along with documentation describing the modelling approach, feature engineering, and evaluation methodology.
Evaluation	Hidden ground truth using ROC-AUC, F1 Score, Precision, and Recall.

The strongest entries will tell a coherent story across both parts.

3 Dataset Description

You are provided with several interconnected CSV files containing approximately 65,000 business travel records.

3.1 Main Trip Data (`public_trip_data.csv`)

This is your primary training dataset. Each row represents a single business trip. Key columns include:

- **Identifiers:** `TripID` (integer)
- **Locations:** `DepartureLocationCountry`, `DepartureLocationCity`, `ArrivalLocationCountry`, `ArrivalLocationCity`
- **Logistics:** `ShippingType`, `ShippingTypeDescription` (e.g., Economy Flight, Train)
- **Business Context:** `Purpose`, `OutOfPolicy`, `EntitiyCode`, `BusinessUnit`
- **Metrics:** `HotelNights`, `NetCosts`
- **Label:** `HighCarbon` (0 or 1)

3.2 Private Test Set (`private_trip_data.csv`)

This contains features for ~22,000 trips used for scoring. **Emissions and labels are hidden.** You must predict `HighCarbon` for these trips.

3.3 Event Logs & Attributes

You also receive process event logs and attribute tables (split into public and private sets).

4 High-Carbon Trip Prediction Challenge

The objective of this challenge is to develop a binary classification model that predicts whether a business trip belongs to the `HighCarbon` class based on the information available before the trip is undertaken.

Participants are encouraged to perform comprehensive exploratory data analysis, engineer informative features, and develop robust machine learning models that generalise well to unseen data.

4.1 Target Label

The training dataset contains a binary target variable named `HighCarbon`, where:

- **`HighCarbon = 1`** indicates a high-carbon business trip.
- **`HighCarbon = 0`** indicates a non-high-carbon business trip.

Your objective is to learn the relationship between the available trip attributes and the target label, and accurately predict the probability of the `HighCarbon` class for the unseen private test set.

4.2 Feature Usage Guidelines

To ensure a fair predictive modelling task, participants must not use variables that directly reveal or are derived from the target label.

The following columns are strictly prohibited as model features:

- `Departure_C02e`
- `Return_C02e`
- `Hotel_C02e`
- `Spend_C02e`
- `TotalC02e`
- `HighCarbon`

All remaining variables—including trip details, travel mode, business context, costs, locations, and the provided process data—may be used for exploratory analysis, feature engineering, and model development.

5 Submission and Evaluation

5.1 Submission Format

Submit a CSV file matching exactly the format of `sample_submission.csv`. It must have two columns:

- `TripID`: Integer identifier matching the private test set.
- `HighCarbon`: Float (predicted probability between 0.0 and 1.0).

5.2 Scoring Metric

Submissions will be evaluated on the hidden private test set using the predicted probabilities for the `HighCarbon` class.

- **Primary Metric: ROC-AUC (Receiver Operating Characteristic – Area Under the Curve)**

ROC-AUC measures how effectively the model distinguishes between high-carbon and low-carbon trips across all possible classification thresholds. A higher ROC-AUC indicates better ranking performance and greater robustness to threshold selection.

- **Supplementary Metrics**

For qualitative evaluation, Precision, Recall, F1 Score, and the Confusion Matrix may also be reported to provide additional insight into the model's classification performance.

The final ranking will be determined using the **ROC-AUC score** computed on the hidden private ground truth.

6 Timeline and Rules

- **No external data:** You may only use the provided datasets.
- **No target engineering:** Do not attempt to reverse-engineer the label thresholds from allowed features.

Good luck, and may the best insights win!